, 059.05

1955

# GLYCIM, a farm-tested model of the soybean crop

Basil Acock[1,2] and V.R. Reddy[1]

Calibration

MDAC 2-8 SP1

---

[1] USDA, ARS, Systems Research Laboratory, Building 007, Room 008, BARC-W 10300 Baltimore Avenue, Beltsville, MD 20705-2350, USA.

[2] Corresponding author

# GLYCIM, a farm-tested model of the soybean crop


## Abstract

Crop models give us a vehicle for delivering detailed scientific knowledge directly to users. Farmers, scientists and policy-makers need a knowledge of plant behavior to answer questions about management options, global climate change, erosion, water quality, etc. GLYCIM was developed as a multi-purpose platform to make available knowledge about the soybean crop. It has a modular structure, and uses mechanistic descriptions of the major system processes to determine whether temperature, carbon, water, or nitrogen limit growth each hour of the day. It incorporates a two-dimensional soil process model, and root growth and symbiotic nitrogen fixation occur preferentially in the most favorable parts of the profile. Carbon is partitioned to maintain a functional equilibrium; if some material is in short supply in the plant, the organ responsible for acquiring that material receives more carbon. GLYCIM has been completely documented and was released to soybean farmers in 1991. To facilitate use of the model, an intuitive graphical interface, WINGLY was developed. With the recent development of cultivar parameter files, both farmers and scientists are using the model for input optimization: selecting cultivars, planting date, and soil type combinations for maximum yield. The farmers also use the model to make in-season decisions on timing of various operations and amount of irrigation. They have reported yield increases of 10 to 29% and irrigation efficiency increases of 400%. The accuracy of GLYCIM's predictions are limited primarily by our incomplete understanding of some plant physiological processes. Research into these processes and development of GLYCIM continues.

## Introduction

Much of our detailed scientific knowledge of plant behavior is unavailable to potential users. It is reported in the special languages of various disciplines, in many different expensive publications that languish on library shelves. At present, science only affects farmers' decisions as that detailed knowledge is used to design field plot experiments, and the results of those experiments are communicated to farmers as general prescriptions for managing their crops under the conditions normally experienced in that area. Crop models give us a vehicle for delivering our detailed knowledge directly to users. Models also have the potential for advising farmers how to respond to unusual conditions, where the general prescriptions break down. However, farmers are not the only potential users. Scientists and policy-makers outside agriculture need a knowledge of plant behavior to answer questions about global climate change, erosion, water quality, etc. Crop models can give them that knowledge in a usable form. GLYCIM was developed as a multi-purpose platform to make available knowledge about the soybean crop.

## Design of GLYCIM

GLYCIM was designed to meet the following criteria: (a) To be capable of incorporating detailed scientific knowledge about crop, soil and atmospheric processes, (b) To be capable of responding to all weather, soil and management variables, (c) To require limited input: a description of the soil, the weather record for the growing season, some cultivar characteristics and management data that could be supplied by a farmer, (d) To predict values for all plant variables commonly observed in experiments, (e) To be able to predict plant response to conditions not previously experienced, e.g. changed global climate, and (f) To be easily modified as new knowledge became available.

To meet these requirements, we decided: (a) To include all mechanisms essential for generating the required output, (b) To model a single, typical plant, (c) To base the model on carbon balance, (d) To calculate potential growth rate and limitations to growth, then use the limiting factor model to determine actual growth (See below for more detail), and (e) To avoid the use of site-specific calibration factors and other factors unrelated to processes and used only to improve model predictions. The rationale for many of these choices is discussed in Whisler et al. (1986).

There are several plant physiological processes for which the mechanisms are unknown, for example, control of the number of branches, timing of flowering, timing of leaf senescence, and control of stomatal aperture. In such instances, mechanisms have been proposed and built into the model that, are plausible to plant physiologists, do not conflict with any of the data in the literature and give reasonable predictions. Even so GLYCIM still contains some relationships that are only empirical.

**Interaction of factors limiting plant growth**

Three models are commonly used to describe the interactions of factors that reduce plant growth, and none of them is perfect. They are the additive, multiplicative and limiting factor models. Suppose that plant attribute Y (yield) is affected by factors A, B and C. The additive model supposes that each factor contributes some fraction of Y. $Y = aA + bB + cC$. where a, b and c are parameters. This is the common regression model widely used in statistics. In this model, Y can have a high value even if one factor is absent; clearly an erroneous result. As the number of factors increases, the fraction of Y that each contributes gets smaller, and the influence of each factor on Y diminishes. In the multiplicative model, $Y = f(A) \times f(B) \times f(C) \times P$, where $f$(factor) is a function of the factor with a value between 0 and 1, and P is potential Y when $f(A) = f(B) = f(C) = 1$. This model works well if only one factor is less than optimal. However if all three factors have a reduced level such that $f(A) = f(B) = f(C) = 0.5$, then $Y = 0.5 \times 0.5 \times 0.5 \times P = 0.125\ P$. This result too is clearly erroneous, giving a value of Y that is much too low, and the error increases with the number of factors in the model. In the limiting factor model, $Y = \mathrm{MIN}\ (f(A)P, f(B)P, f(C)P)$. In this model, as A increases at some fixed level of B and C, Y also increases up to some point then suddenly becomes constant as B or C limit the response. In fact, Y does not change in this abrupt manner in experimental data; there is a more gradual approach to the plateau. Nevertheless, the limiting factor model gives the best results when the number of factors under consideration is large. It acknowledges that no factor can replace any other factor, and that all factors are required in balanced amounts. It also facilitates the development of models from controlled-environment experiments, which are almost exclusively single-factor experiments.

**Structure of GLYCIM**

When GLYCIM was first completed and documented (Acock et al., 1983), the authors expressed the hope that other domain specialists would improve those parts of the model in their domains. Eventually it became apparent that the equations relevant to any one expert were scattered throughout the code and, without an intimate knowledge of the code, the experts could not find them. Clearly, the equations relevant to each expert had to be gathered together in easily identifiable units: the model had to have a modular structure. Because we were using a procedural language, we based our structure on processes in the plant, soil and atmosphere, and devised a generic modular structure in collaboration with other modelers (Acock and Reynolds, 1989; Reynolds et al., 1989). The code of GLYCIM was rearranged to fit the new structure (Fig. 1) and documented again (Acock and Trent, 1991). Since that time, GLYCIM has continued to evolve and this paper describes version 95. Also since 1991, much thought has been given to defining modularity. It involves the independence and testability of modules as well as gathering together related equations (Reynolds et al., 1993). The present trend is to base modules on physical objects in the system being modeled rather than on processes (Acock and Reddy, 1996), and this is being considered for future versions of GLYCIM.

**Timestep**

GLYCIM operates with an hourly step to better simulate plant water stress. Even healthy plants show signs of water stress around noon on sunny days, and they can compensate fully for that. However, the longer the period of stress, the less able the plant is to compensate. The hourly step allows us to determine how long the model plant spends in stress and whether the loss of growth is irreversible.

## Input

Weather data At a minimum, GLYCIM requires daily data on solar radiation, maximum and minimum air temperatures and rainfall plus irrigation water applied. It can use data on windrun and humidity if available. From these GLYCIM calculates hourly data, but it can also use hourly observations.

Soil data For each soil horizon, GLYCIM requires depth, texture, bulk density, saturated hydraulic conductivity and three or more points on the pF (water content v. water potential) curve. Residual nitrogen content, organic matter, fertilizer nitrogen applied, and deep soil temperature are also needed. Initial water contents may be input if known, otherwise field capacity is assumed at emergence.

Initialization data Additional necessary inputs are emergence date, plant population, row spacing, row orientation, latitude of the site, carbon dioxide concentration in the air, and various cultivar characteristics

## Housekeeping module

MAIN calls the other modules. SOILIN is called once at the start of each simulation, WEATHER and LYTINT are called once each day, and all other modules once each hour.

## Atmospheric process modules

WEATHER uses meteorological data and celestial geometry to calculate day length, effective photoperiod for soybean, cloud cover, and hourly values of some environmental variables, including air temperature, vapor pressure deficit, the proportion of diffuse radiation, and photosynthetically-active radiation. A half-sine wave is used to describe the diurnal distribution of radiation. A half-sine wave is also used for temperature, with its peak shifted after noon by an amount dependent on radiation integral and maximum temperature for the day. At night the temperature is assumed to fall logarithmically (Fig. 2). Direct and diffuse radiation are handled separately because diffuse radiation is intercepted more efficiently by crops with incomplete cover and also contains 33% more photosynthetically-active radiation than direct radiation . Diffuse radiation is an important component of total radiation in areas with frequent cloud cover. WEATHER estimates cloud cover and hence the ratio of diffuse to direct radiation by comparing actual radiation for the day with expected radiation for that latitude and day of year (Fig. 3). The proportion of diffuse radiation expected for a cloudless day is calculated from path-length through the atmosphere for that latitude and day of year. The data generated in WEATHER are used throughout the model.

LYTINT (Light interception) calculates hourly values of total and photosynthetically-active radiation intercepted by the crop canopy. It uses a hedgerow model of light interception prior to canopy closure. A circular cross-section of hedgerow is assumed but other shapes are easily substituted. Celestial geometry and row orientation are used to calculate hourly the length of shadow cast by a hedgerow on the soil surface (Fig. 4). From this are calculated the percentages of direct and diffuse radiation falling on the soil and the plant canopy. Effective leaf area index within the volume of the canopy is then used to calculate radiation interception by the canopy (Fig. 4). The radiation interception data are used in modules PNET and TRANSP.

**Soil process modules**

SOILIN (Soil initialization)uses data on the characteristics and initial conditions of soil in the various horizons of the profile to set initial conditions in each cell of a representative modeled slab of soil. The slab extends from the base of the plants in one row to the base of the plants in the next, or to the midrow if fertilizer, irrigation water, etc are applied uniformly on both sides of the row (Fig. 5). The slab is 1 cm thick in the direction of the row. The number and other dimensions of the cells are set by the operator. SOILIN calls THERMS.

THERMS (Soil thermal properties) calculates the heat capacity and thermal conductivity of the soil in each horizon at several temperatures and water contents using the equations of Van Wijk and colleagues (1966). THERMS calls MULREG.

MULREG (Multiple regression) executes a regression analysis on the data for each horizon and returns regression coefficients for rapidly calculating soil heat capacity and thermal conductivity in TMPSOL. MULREG uses CORMAT and INVERT to manipulate the matrices.

SOILEN (Soil environment) updates volumetric water content, water potential, hydraulic conductivity, oxygen concentration, temperature, and concentrations of ammonium and nitrate in each cell in the soil profile. SOILEN is a modified form of RHIZOS, the two-dimensional soil process model used in GOSSYM (Baker et al., 1983). SOILEN provides the plant model with: (a) nitrogen availability in the soil for calculating nitrogen uptake, (b) soil water potential for calculating leaf water potential, plant turgor level and reductions in transpiration, photosynthesis and growth induced by water stress, (c) root sink strength for carbon and nitrogen, and (d) soil environmental conditions affecting root and nodule growth, respiration rates, and nitrogen fixation. The plant model provides SOILEN with: (a) water and nitrogen uptake, and (b) soil surface shading. SOILEN is a shell module that only calls EVAPOR, TMPSOL, SURFIR, GRAFLO, UPTAKE, CAPFLO, NITRIF and SOLOXY. All these modules perform calculations for each cell in the model profile.

EVAPOR (Evaporation of soil water) calculates actual water evaporation from the exposed surface soil cells and the amount of radiation available for heating the soil. Actual evaporation rate is the potential rate calculated with Penman's equation (Penman, 1963) or the rate at which water can move to the soil surface, whichever is least (Rowse and Stone, 1978).

TMPSOL (Soil temperature) calculates a radiation balance for the soil surface exposed between plant rows, heat flux into the soil, and temperature of each soil cell. Radiation not intercepted by the plant canopy is assumed to fall on the soil, There the energy may heat the soil, evaporate water, or be lost by reradiation or convection. Actual evaporation rate has been calculated already in EVAPOR and can be subtracted. To solve the reradiation equation accurately, we need to know soil surface temperature. However the reradiation equation can be approximated quite well by a function of the air/surface temperature difference. Using this equation in GLYCIM, we are able to write a set of simultaneous equations for the various forms of energy dissipation, and solve them without knowing surface temperature (Fig. 6). Conduction is assumed to be a negligible form of energy loss from exposed soil cells, but is the only form of loss from cells under the canopy. Heat flux in the soil and hence soil temperatures throughout the profile are calculated according to the method of Van Wijk (1966).

SURFIR (Surface irrigation) calculates the infiltration of water into the soil from flood, furrow and sprinkler irrigation when water ponds on the whole or part of the surface. A Green-Ampt approach is used to simulate infiltration into layered soil (Pachepsky and Timlin, 1995).

GRAFLO (Gravity flow) calculates the flow of water and nitrate down the profile. The top cell of each column is filled to apparent field capacity before passing any remaining water to the next cell down the column. Apparent field capacity is equal to the soil water content predicted from the Green-Ampt formulae when ponding occurs. Between ponding events, apparent field capacity is equal to true field capacity. It is assumed that runoff does not occur.

UPTAKE calculates changes in water and nitrate content of each soil cell resulting from root water uptake. Water uptake is calculated in TRADE and nitrate uptake in NITRO, but UPTAKE actually changes the state of the soil in the model.

<u>CAPFLO</u> (Capillary flow) calculates the capillary flow of water and nitrate in all directions within the modeled soil slab. Darcy's equation is used to move water along gradients of soil water potential. Hydraulic conductivity is a function of water content based on saturated hydraulic conductivity. This redistribution of water is required because water taken up by roots and evaporated from the soil surface causes water potential gradients. Nitrate is assumed to all be in solution and to move with the water.

<u>NITRIF</u> (Nitrification) calculates the nitrification of ammonium and mineralization of organic matter, using the model of Kafkafi et al (1978). The conversions of organic matter to ammonium and ammonium to nitrate are functions of soil temperature and soil water content. It is assumed that only nitrate is taken up by the plants.

<u>SOLOXY</u> (Soil oxygen) calculates the oxygen concentration in the root zone. It accounts for the size and tortuosity of pores, oxygen diffusion from the surface and oxygen consumption by roots, nodules and organic matter decomposition. Respiration rates are calculated in ACTGRO for roots, NODULE for nodules and NITRIF for the microbes decomposing organic matter.

**Plant physiology modules**

<u>PHEN</u> (Phenology) calculates the vegetative and reproductive stages of development. The rate of appearance of leaves on the mainstem and branches is a well-documented function of temperature. The timing of reproductive stages, though often measured, is less well understood and involves both temperature and daylength. GLYCIM recognizes one more reproductive stage than those enumerated by Fehr and Caviness (1977); R0 is floral initiation. Based on data of Garner and Allard (1930), we assume that the rate of progress between unifoliolate expansion (V1) and R0 is a function of maturity group and daylength that differs for increasing and decreasing daylengths (Fig. 7). R0 to R1 is a function of temperature. R1 to R3 uses a function very like that shown in Fig. 7. R3 to R4 is a function of daylength independent of

maturity group. Between R4 and R7, reproductive stage is a function of percent seedfill for the oldest and largest seed on the plant. This is arbitrary, but R5 and R6 are not important stages in the model. When the oldest seed reaches maximum size, monocarpic senescence is assumed to start. One layer of leaves, i.e. a mainstem leaf and branch leaves of the same age, falls off each day subsequently until R8 when the plant has completely defoliated.

PNET (Net photosynthesis) uses single-leaf photosynthetic characteristics to calculate crop canopy characteristics and canopy gross photosynthetic rate. Gross photosynthetic rate is given a hyperbolic dependence on both light intercepted by the crop and carbon dioxide concentration, and it is limited by low temperatures. Photorespiration rate is a function of temperature and the ratio of oxygen and carbon dioxide concentrations. Calculation of the maintenance respiration rate, required to repair damage to proteins and membranes, uses the ideas and data of Penning DeVries (1975). It is assumed that only leaf tissue requires appreciable maintenance and that maintenance respiration rate has a Q10 of 2. These respiration rates are subtracted from gross photosynthetic rate to get net photosynthetic rate, which is corrected for stomatal conductance (determined in TRADE). From this are calculated the net carbon fixation rate and the rate of carbon translocation out of the leaves. Growth respiration is charged against the translocated carbon in ACTGRO, where actual growth of organs is computed. A detailed description of this module is presented in Acock (1991). Net photosynthesis is also calculated for the lowest layer of leaves in the canopy to see if they are "parasitic" on the plant and should be abscised.

POTGRO (Potential growth) calculates potential rates of growth for all organs on the plant at the current air and soil temperatures assuming that carbon, water, and nutrients are plentiful. POTGRO is a shell module that only calls SHTGRO and RUTGRO.

SHTGRO (Potential shoot growth) calculates potential growth rates of shoot organs at the current air temperature assuming that carbon, water, and nutrients are plentiful. Stems, petioles, leaves, pods and seeds are all tracked individually. Organ expansion rate is calculated first and then the maximum and minimum rates of dry weight gain that could accompany the expansion. Some organs have more flexibility in their dry weight per unit volume than others; leaves are quite flexible, seeds are not.

RUTGRO (Potential root growth) calculates potential rates of root growth in each soil cell for the current temperatures, water contents and oxygen concentrations assuming that carbon, and nutrients are plentiful. Potential root growth in the model is high and is usually limited by availability of carbon from the shoot, as determined in TRADE.

PARTIT (Partitioning) calculates an initial partitioning of carbon to various organs based on priorities that change with stage of growth, and assuming that water and nitrogen are plentiful. The priorities are as follows, where "X = Z" indicates that X and Z have equal priority and share any available carbon, and "X > Z" indicates that X has absolute priority over Z. Prior to R1: nodules > vegetative organs. R1 to R3: nodule growth in anticipation of fruiting > normal nodule activity > vegetative organs. R3 to R4: pods = storage of carbon equivalent to that required for potential seeds = nodules > vegetative organs. After R4: seeds = pods = nodules > vegetative organs. Note that GLYCIM grows nodules in anticipation of the heavy demand for nitrogen during fruiting (pod and seed growth). Also, while it is adding pods (R3 to R4), GLYCIM stores carbon equivalent to that required for potential seed growth. These artificial devices simulate observed plant behavior but provide no explanation for it. Without them, nitrogen limits the addition of pods at R3. Then, by R4, nodule activity has restored nitrogen content and the plant is carrying more pods than it can subsequently fill.

WATERS (Plant water relations) calculates any limitations to growth caused by low soil water availability. WATERS is a shell module that only calls TRANSP and TRADE.

TRANSP (Transpiration) calculates potential transpiration rate from the crop using Penman's (1963) equation applied to that portion of the field covered with crop.

TRADE is a unique module for handling plant water relations. It maintains a functional balance between root and shoot by growing roots, as necessary and possible, to meet transpiration demand. It calculates leaf water potential, shoot and root growth rates, stomatal conductance, and actual transpiration rate. It does this by calculating water uptake by roots for a number of key shoot water potentials and comparing these with potential transpiration rate. On Fig. 8, point A represents zero water uptake when leaf water potential equals soil water potential. Point B is leaf water potential one hour ago, C is the threshold leaf water potential at which leaf expansion is just prevented and all the carbon available goes to grow new roots, D is the point at which stomatal closure starts, and is the leaf water potential at which leaf turgor pressure is 0.2 MPa (2 bars). If potential transpiration rate is between the root water uptake rates corresponding to A and D, the plant is assumed to adjust leaf water potential and carbon partitioning to meet the demand; actual transpiration rate equals potential. For potential transpiration rates above the root water uptake rate at D, the stomata close progressively and actual transpiration falls below potential. Two classes of roots are recognized: suberised and unsuberised. The unsuberised roots have less resistance to water uptake, so new root growth can markedly affect water uptake rate. This is why the slope AB is steeper than AC. The locations A, B, C and D can all change from one hour to the next.

NUTRTS (Nutrients) calculates any limitations to growth caused by low nutrient availability. It is structured to deal with any nutrients but actually only considers nitrogen. NUTRTS is a shell module that only calls NITRO and NODULE.

NITRO (Nitrogen) calculates the supply of and demand for nitrogen in the whole plant including the potential new growth. It determines a stress factor that is used to partition carbon to the nodules, limit photosynthesis and shoot growth, and control leaf and petiole abscission. Soybean tissue can be synthesized with nitrogen contents as low as 40% of normal. Since soybeans are legumes, nitrogen availability rarely limits shoot growth directly, but it sometimes limits growth through its effect on photosynthesis.

NODULE calculates nodule growth, nitrogen fixing activity and death in each soil cell. It is assumed that nitrogen stress causes carbon to be allocated to grow nodules and fix nitrogen. Thus GLYCIM does not grow nodules until fertilizer nitrogen in the soil has been exhausted. The new nodules have to be 3 days old before they start fixing nitrogen and they take time to reach a useful size, so the model plants often experience a short period of nitrogen stress like field plants. Nodule growth rate and nitrogen fixation rate are functions of soil temperature and soil oxygen concentration. Carbon allocated to the nodules is used to fix nitrogen, grow existing nodules and initiate new nodules, in that order of priority. Carbon is used preferentially in those soil cells with the best conditions for nodule activity.

ACTGRO (Actual growth) Uses the initial carbon allocation plus any limitations caused by water stress or nitrogen stress to calculate the actual growth in size and dry weight of all organs on the plant. Root growth occurs in the soil cells where conditions are most favorable and is shared with adjacent cells. ACTGRO initiates branches, pods and seeds according to the plant's ability to support them. A simplified diagram of the overall carbon partitioning scheme in GLYCIM is shown in Fig. 9. GLYCIM applies the limiting factor model to each organ or class of organ in turn depending on their position in priority for that stage of growth. Certain organs grow while others do not; the available carbon is not proportioned among growing organs as in most models. This enables the model plant to maintain functional equilibrium between carbon, water and nitrogen availability, by sending carbon preferentially to the organ responsible for obtaining the material that is limiting growth. For example, roots grow preferentially during water stress. Sizes of organs on the plant are determined primarily by temperature and water availability and weights are determined primarily by carbon and nitrogen availability during the period of organ growth.

TISLOS (Tissue loss) removes from the plant, tissue that is lost for any reason. Tissue, usually whole organs, is marked for removal (abscission) in other parts of the model but the state of the plant is actually updated in this module.

**Validation and future development**

GLYCIM was developed from data for the Forrest cultivar grown in controlled-environment and field experiments in Mississippi. It has been compared to many field datasets, principally from the south-east US and improved as we have learned more. Fig. 10 is an example of observed and predicted data for a farm in Mississippi. We now believe that our present understanding of plant physiology does not allow us to construct crop models that will accurately predict yields with all datasets. As stated earlier, the mechanisms controlling time of flowering, number of branches, timing of leaf senescence, and stomatal conductance are still unknown. The Systems Research Lab (SRL) is continuing work to elucidate control of flowering (Acock et al., 1994; Acock and Acock, 1995), and is developing a detailed model of leaf physiology including stomatal functioning (Pachepsky et al, 1995). Occasionally soybeans grow more slowly after germination than we expect from the air temperature. SRL also has a research program to determine the reason for this so that the appropriate mechanism can be incorporated in GLYCIM. The RHIZOS code in GLYCIM is being replaced with 2DSOIL (Timlin et al., 1995) because the latter tracks water and solute fluxes more accurately and includes more soil chemistry.

## Model application and impact

GLYCIM was released to soybean farmers in 1991. To facilitate use of the model, an intuitive graphical interface, WINGLY was developed (Amonson et al., 1995). Actual yield predictions were poor at first, but the farmers learned about the physiology of their crop, and found that they could use yield differentials to make management decisions. The use of maturity grouping to describe cultivar characteristics proved inadequate and recently cultivar parameter files were developed (Reddy et al., 1995). Now yield predictions are more accurate and both farmers and scientists are using the model for input optimization, selecting cultivars, planting date, and soil type combinations for maximum yield. The farmers also use the model to make in-season decisions on timing of various operations and amount of irrigation. They have reported yield increases of 10 to 29% and irrigation efficiency increases of 400% (Remy, 1994). The ability of GLYCIM to simulate 20 years of soybean yields for all counties in Iowa has been demonstrated and it is now being used to examine possible effects of global climate change on soybean production in Iowa (Haskett et al., 1995).

## Technical

GLYCIM is written in Fortran77 and has been run on many different PCs, Macintoshes and a Cray. On a 386 with math co-processor, or on a 486, GLYCIM typically simulates one season in about 10 minutes. Hard copy of the 1991 documentation is available from the authors, and the latest compiled version of the code can be downloaded from the Web site http://www.arsusda.gov

## Acknowledgements

Most scientific knowledge results from a steady accretion of data and occassional flashes of insight into what it all means. GLYCIM is an accretion of hypotheses. The authors of this paper are aware that GLYCIM is built on the ideas of countless others; modelers are generous with their thoughts and their code. There may be others, but we are especially aware of our debt to Don Baker, Jerry Lambert, Ken Boote, Tony Trent, Frank Whisler and Harry Hodges.

## References


Acock, B. 1991. Modeling canopy photosynthetic response to carbon dioxide, light interception, temperature, and leaf traits. pp 41-55. In CSSA Special Publication number 19, Modeling crop photosynthesis - from biochemistry to canopy.

Acock, B. and V. R. Reddy. 1996. An object-oriented structure for crop models. Ecological modelling. (accepted for publication).

Acock, B., V. R. Reddy, F. D. Whisler, D. N. Baker, J. M. McKinion, H. F. Hodges and K. J. Boote. 1983. The soybean crop simulator GLYCIM. Number 002 in series: Responses of vegetation to carbon dioxide. U.S. Dept. of Energy and Dept. of Agriculture.

Acock, B. and J. F. Reynolds. 1989. The rationale for adopting a generic modular structure for crop simulators. Acta Hort., 248:391-396.

Acock, B. and A. Trent. 1991. The soybean crop simulator GLYCIM: documentation for the modular version 91. Number 017 in series: Response of vegetation to carbon dioxide. U.S. Dept. of Energy and Dept. of Agriculture.

Acock, M. C., J. A. Bunce and B. Acock. 1994. Effect of changing daylength on flower initiation and development in two soybean cultivars. Biotronics, 23:93-104.

Acock, M. C. and B. Acock. 1995. Photoperiod sensitivity during soybean flower development. Biotronics, 24:000-000 (In press).

Baker, D. N., J. R. Lambert and J. M. McKinion. 1983. GOSSYM: a simulator of cotton crop growth and yield. South Carolina Agric. Exp. Stn. Tech. Bull., 1089.

Fehr, W. R., and C. E. Caviness,. 1977. Stages of soybean development. Iowa Agric. Exp. Stn. Spec. Rep. 80.

Garner, W. W. and H. A. Allard. 1930. Photoperiodic response of soybeans in relation to temperature and other environmental factors. J. Agric. Res. 41:719-735.

Haskett, J. D., Ya. A. Pachepsky and B. Acock. 1995. Simulation of soybean response to changes in $CO_2$ and the effects of climate change. Agron. J., (In press).

Kafkafi, W., ~~B.~~ D. Bar-Yosef and A. Hadas. 1978. Fertilization decision model - a synthesis of soil and plant parameters in a computerized program. Soil Sci., 125:261-268.

Pachepsky, L. B., J. D. Haskett and B. Acock. 1995. A two-dimensional model of leaf gas exchange with special reference to leaf anatomy. J. of Biogeography, 22 (In press).

Pachepsky, Ya. A. and D. J. Timlin. 1995. Infiltration into layered soil covered with a depositional seal: a Green-Ampt approach. Trans. ASAE, 37 (In press).

Penman, H. L. 1963. Vegetation and hydrology. C.A.B. Tech. Comm. 53. Commonwealth Agricultural Bureaux, Harpenden, Hertfordshire, U.K.

Penning de Vries, F. W. T. 1975. The cost of maintenance processes in plant cells. Ann. Bot., 39:77-92.

Reddy, V. R., B. Acock and F. D. Whisler. 1995. Crop management and input optimization with GLYCIM: differing cultivars. Computers and Electronics in Agriculture, (In press).

Remy, K. 1994. GLYCIM model proves its worth. Research Highlights, MAFES Pub., 57. Mississippi State, MS.

Reynolds, J. F., B. Acock, R. L. Dougherty and J. D. Tenhunen. 1989. A modular structure for plant growth simulation models. pp. 123-134 . In Pereira, J. S. and Landsberg, J. J. (eds.) Biomass production by fast-growing trees. Kluwer Academic, Boston, MA.

Reynolds, J. F., B. Acock and R. Whitney. 1993. Linking CO2 experiments and modelling. pp. 93-106. In Schulze, E. D. and Mooney, H. A. (eds.) Design and execution of experiments on CO2 enrichment. Commission of the European Communities: Ecosystem Research Report 6.

Rowse, H. R. and D. A. Stone. 1978. Simulation of the water distribution in soil: I. measurement of soil hydraulic properties and the model for an uncropped soil. Plant and Soil 49:517-531.

Timlin, J. D., Ya. A. Pachepsky and B. Acock. 1995. A design for a modular generic soil simulator to interface with plant models. Agron. J., (In press).

Amonson C. D. P., A. Trent, B. Acock and V. R. Reddy. 1995. WINGLY, a soybean simulation for windows: users manual. Univ. Idaho, Moscow, ID. and USDA, ARS Systems Research Lab., Beltsville, MD.

Van Wijk, W. R. (ed.). 1966. Physics of plant environment. North Holland Pub Co., Amsterdam.

Whisler, F. D., B. Acock, D. N. Baker, R. E. Fye, H. F. Hodges, J. R. Lambert, H. E., Lemmon, J. M. McKinion and V. R. Reddy. 1986. Crop simulation models in agronomic systems. Adv. Agron. 40:141-208.

## List of captions

Fig. 1. The principal modules in GLYCIM 95.

Fig. 2. Diurnal variation in temperature as measured (points for 2nd July, 1994, Mississippi Delta) and simulated in GLYCIM (continuous line). Clearly the weakest part of the model is the empirical equation estimating how long after noon the temperature peak occurs.

Fig. 3. The relationship between cloud cover and diffuse and direct radiation, and the methods used to calculate them. The proportion of diffuse radiation from a cloudless sky is exaggerated to show the principle.

Fig. 4. Radiation interception by hedgerows prior to canopy closure. For calculating Beer's law light interception by the hedgerow, effective leaf area index = leaf area/canopy area.

Fig. 5. The modeled soil profile. Each cell is 1 cm thick. Other dimensions are chosen by the operator.

Fig. 6. Energy balance at the soil surface.

Fig. 7. Time between unifoliolate expansion and floral initiation for maturity groups 4 and 8 in natural (changing) daylengths.

Fig. 8. Hypothetical relationship between leaf water potential ($\Psi$), leaf turgor pressure (P), and potential and actual transpiration rates.

Fig. 9. Simplified diagram of carbon partitioning in GLYCIM. The diabolos represent valves controlling the flow of carbon and dashed lines connect them to the controlling factor. Uses for carbon are listed in priority order. If the plant is unable to use the carbon it reduces photosynthetic rate.

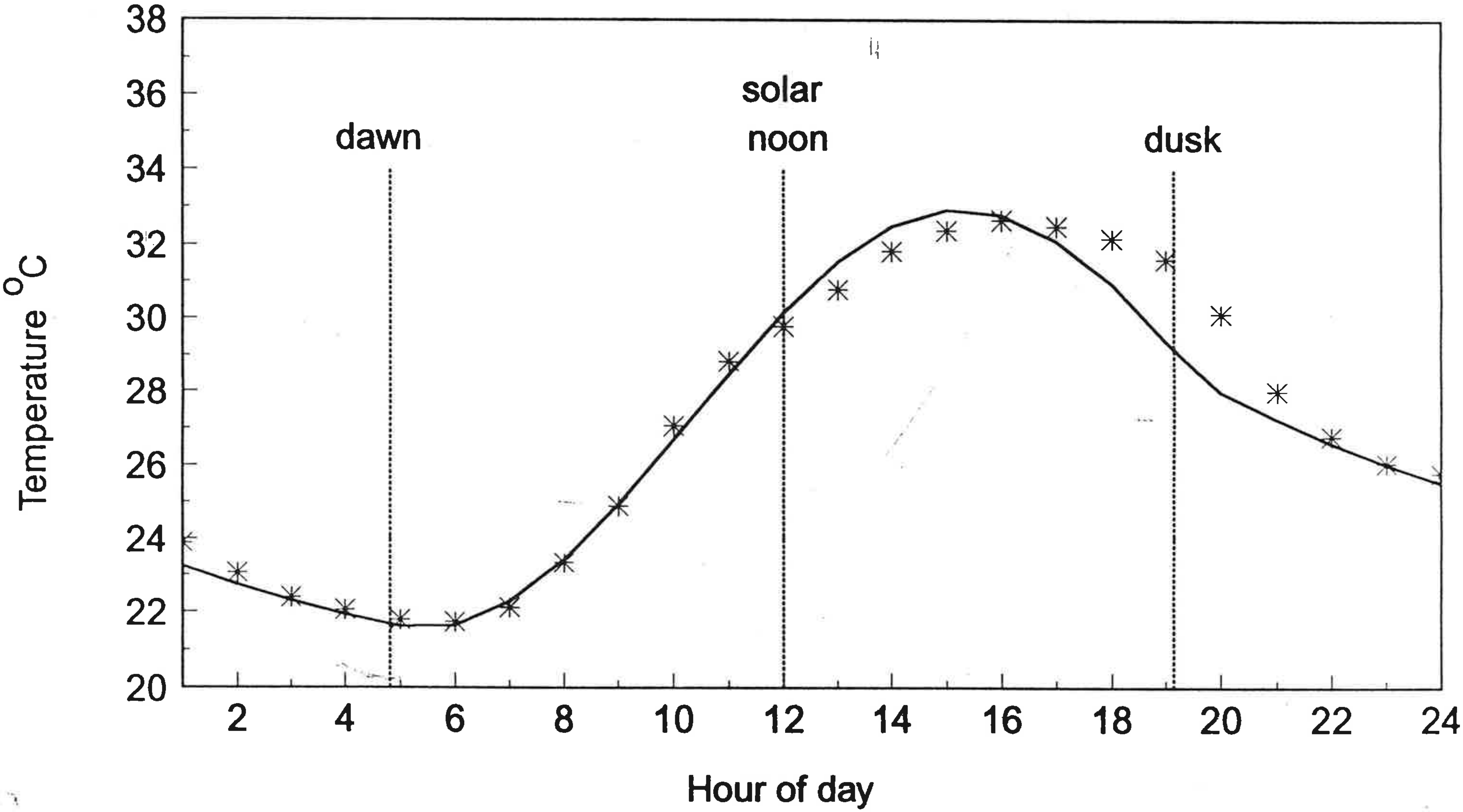
Fig. 10. Plant height, vegetative and reproductive stages observed for cultivar NKS-5960 grown in Dundee loam soil in the Mississippi Delta (points) and simulated with GLYCIM (continuous line). Observed yield was 2379 kg $ha^{-1}$, simulated was 2620 kg $ha^{-1}$.

**Housekeeping module**
MAIN calls the other modules.

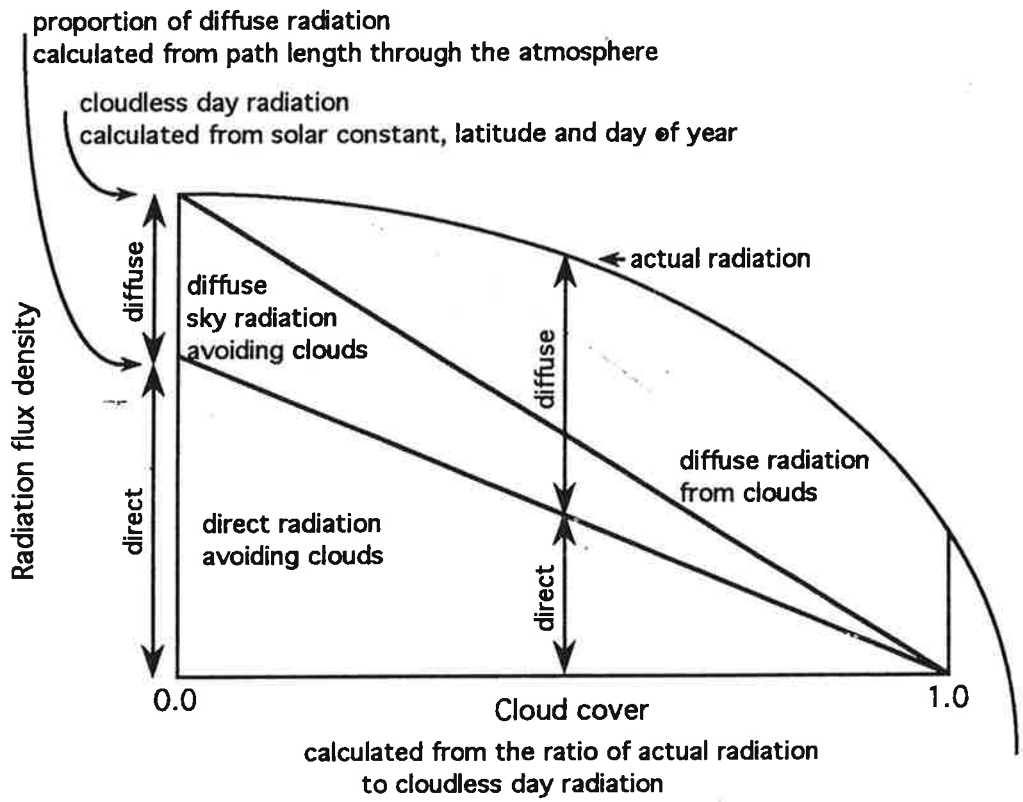
**Atmospheric process modules**
WEATHER calculates hourly values of weather variables
LYTINT (Light interception)calculates total radiation and PAR intercepted by the crop.

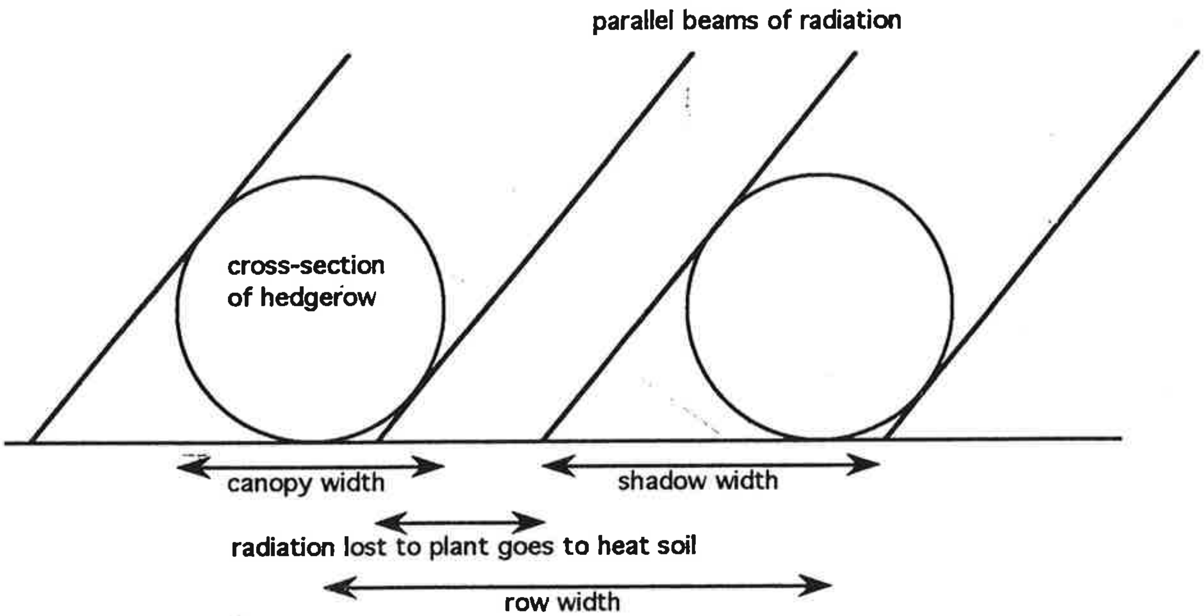
**Soil process modules**
SOILIN (Soil initialization) initializes soil conditions in the model profile.
SOILEN (Soil environment) updates soil environmental conditions.
  EVAPOR (Evaporation of soil water) calculates water evaporated from soil surface.
  TMPSOL (Soil temperature) calculates soil temperature.
  SURFIR (Surface irrigation) calculates infiltration from flood, furrow and sprinkler.
  GRAFLO (Gravity flow) calculates the flow of water and nitrate down the profile after rain.
  UPTAKE calculates changes in water and nitrate content of soil.
  CAPFLO (Capillary flow) calculates capillary flow of water and nitrate.
  NITRIF (Nitrification) calculates the nitrification of ammonium and organic matter.
  SOLOXY (Soil oxygen) calculates the oxygen concentration in the root zone.

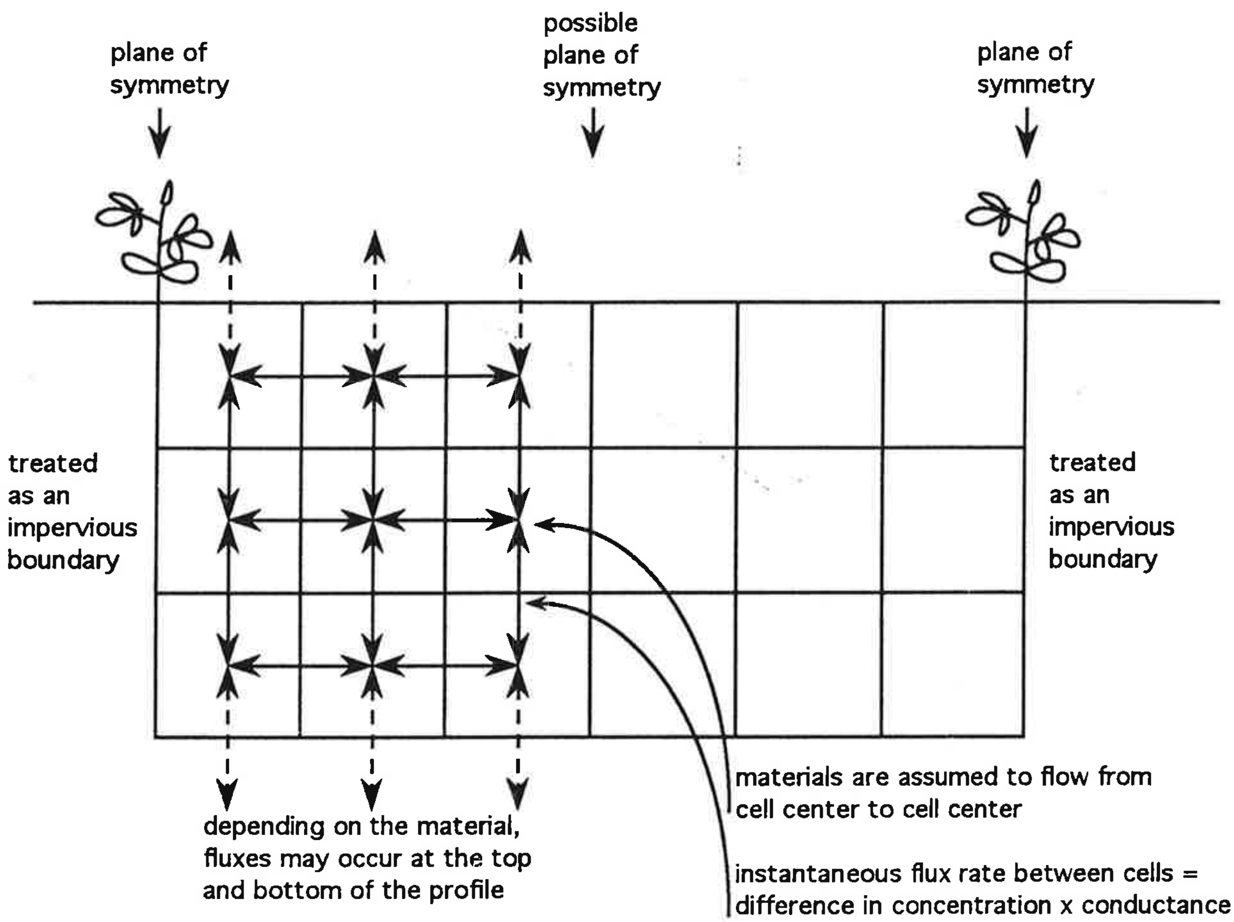
**Plant physiology modules**
PHEN (Phenology) calculates the vegetative and reproductive stages of development.
PNET (Net photosynthesis) calculates crop canopy photosynthetic rate.
POTGRO (Potential growth) calculates potential rates of growth for all organs.
  SHTGRO (Potential shoot growth) calculates potential growth rates of shoot organs.
  RUTGRO (Potential root growth) calculates potential rates of root growth in each soil cell.
PARTIT (Partitioning) calculates an initial partitioning of carbon to various organs.
WATERS (Plant water relations) calculates limitations caused by low soil water availability.
  TRANSP (Transpiration) calculates potential transpiration rate
  TRADE handles plant water relations.
NUTRTS (Nutrients) calculates any limitations to growth caused by low nutrient availability.
  NITRO (Nitrogen) calculates the nitrogen supply and demand.
  NODULE calculates growth, nitrogen fixing activity and death of nodules.
ACTGRO (Actual growth) calculates the actual growth of all organs on the plant.
TISLOS (Tissue loss) removes from the plant, tissue that is lost for any reason.

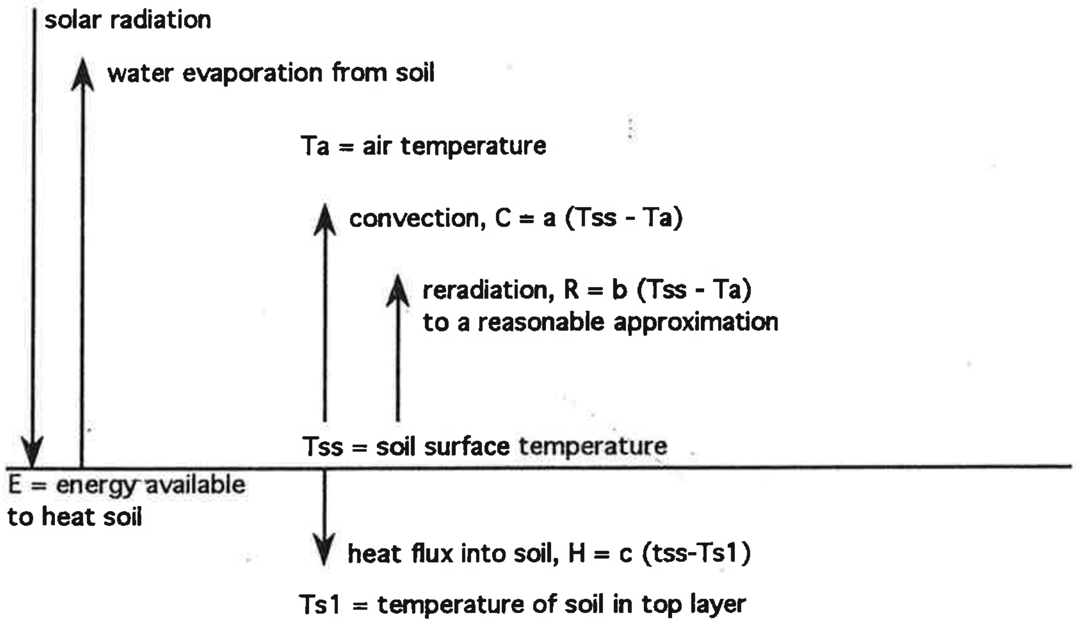


Temperature °C
38
36
34
32
30
28
26
24
22
20
dawn
solar
noon
dusk
2
4
6
8
10
12
14
16
18
20
22
24
Hour of day


proportion of diffuse radiation
calculated from path length through the atmosphere
cloudless day radiation
calculated from solar constant, latitude and day of year
actual radiation
diffuse
diffuse
sky radiation
avoiding clouds
diffuse
diffuse radiation
from clouds
direct
direct radiation
avoiding clouds
direct
Radiation flux density
0.0
1.0
Cloud cover
calculated from the ratio of actual radiation
to cloudless day radiation

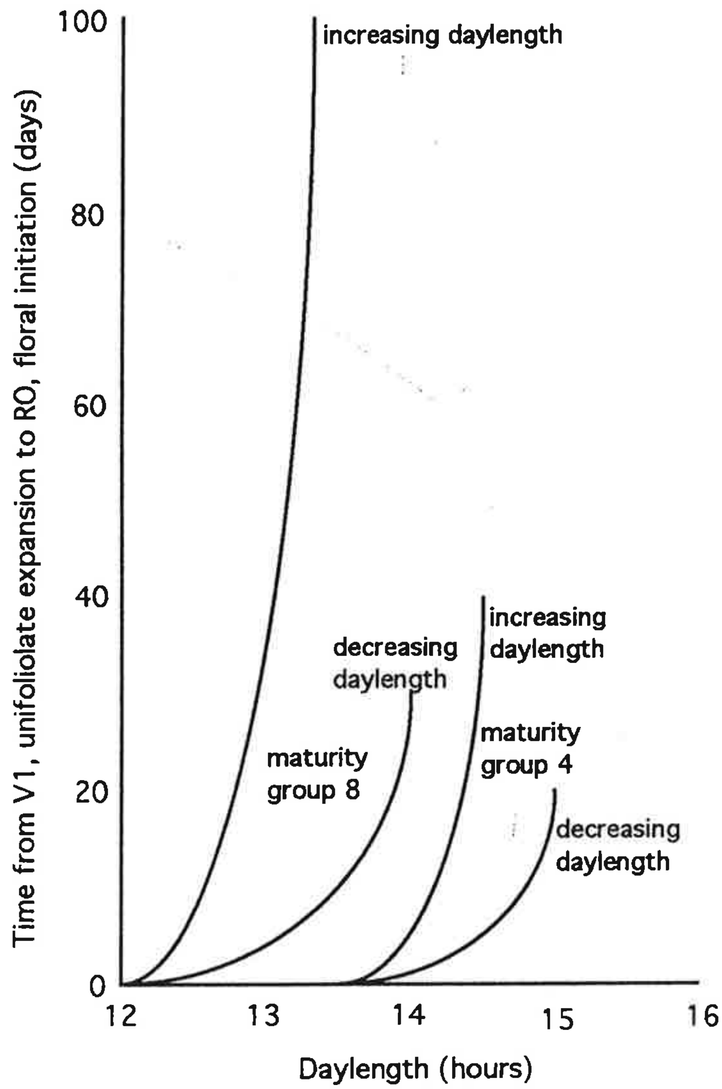

parallel beams of radiation
cross-section
of hedgerow
canopy width
shadow width
radiation lost to plant goes to heat soil
row width


plane of symmetry
possible plane of symmetry
plane of symmetry
treated as an impervious boundary
treated as an impervious boundary
depending on the material, fluxes may occur at the top and bottom of the profile
materials are assumed to flow from cell center to cell center
instantaneous flux rate between cells = difference in concentration x conductance

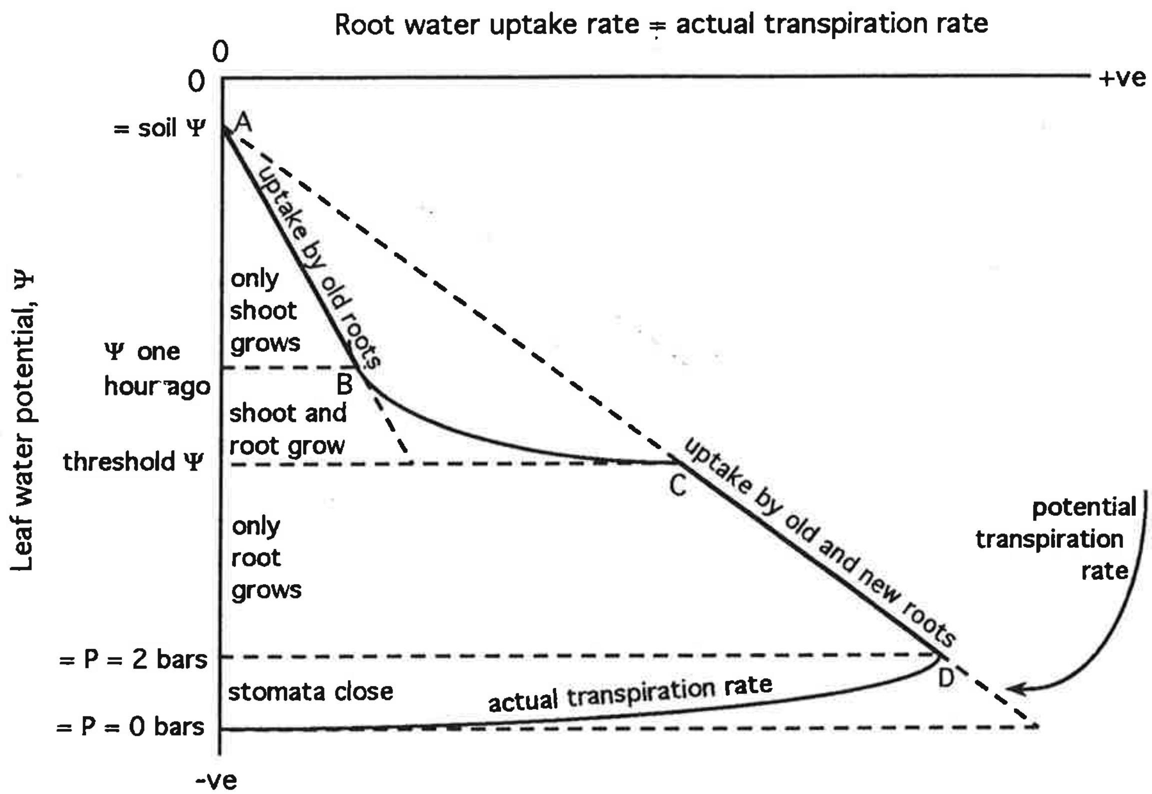



by combining, rearranging and substituting these equations, Tss can be eliminated:

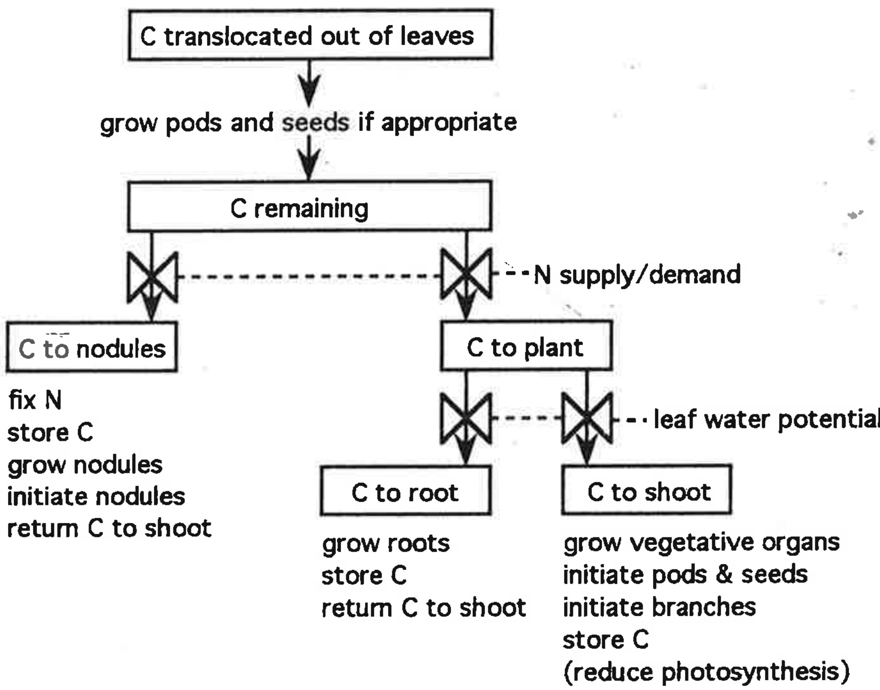
$H = c\ ((E/(a + b)) = Ta + Ts1)/(1 + (c(a + b)))$

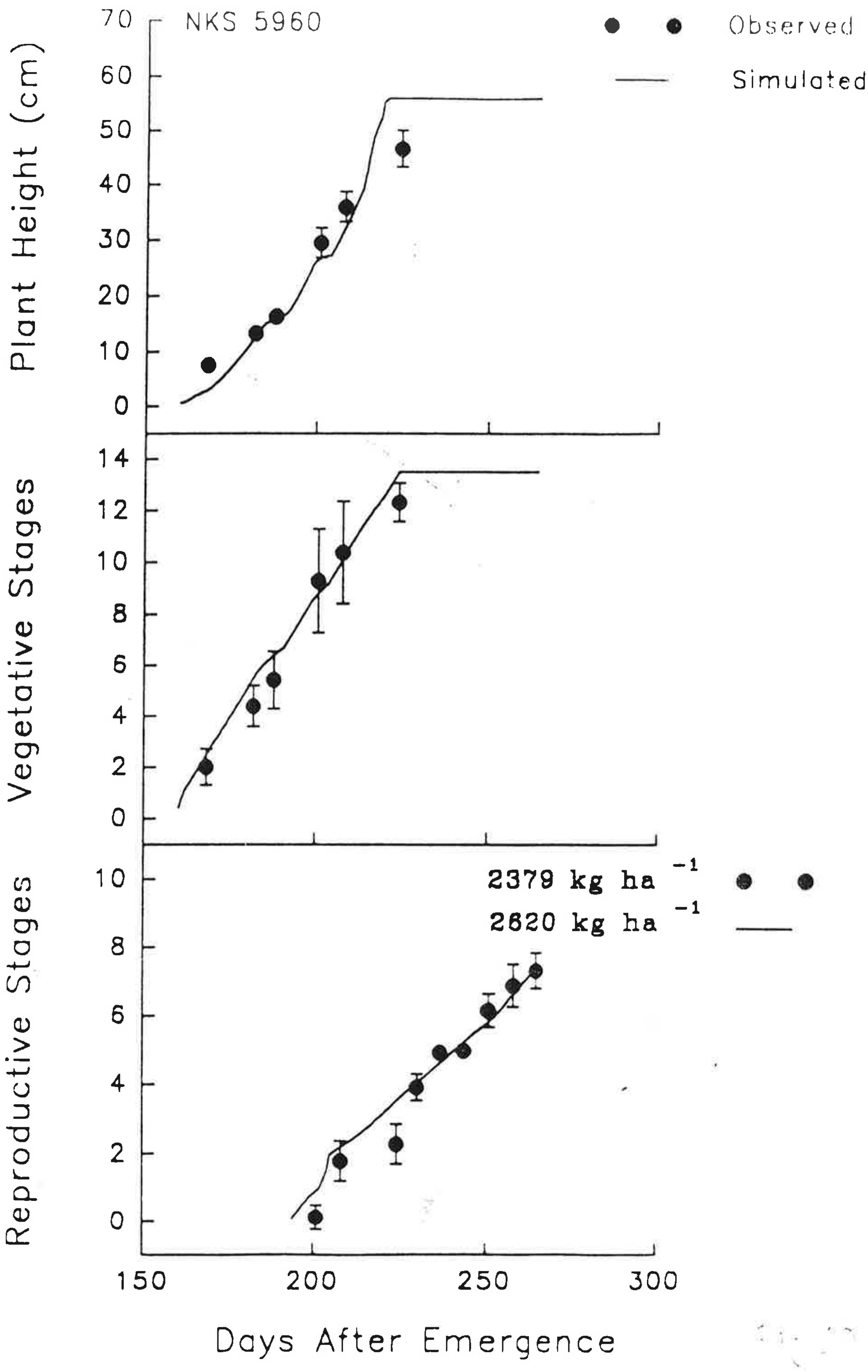


100
80
60
40
20
0
Time from V1, unifoliolate expansion to R0, floral initiation (days)
increasing daylength
decreasing daylength
maturity group 8
increasing daylength
maturity group 4
decreasing daylength
12
13
14
15
16
Daylength (hours)



Root water uptake rate = actual transpiration rate
0
0
+ve
= soil Ψ
A
uptake by old roots
only
shoot
grows
Ψ one
hour ago
B
shoot and
root grow
threshold Ψ
C
uptake by old and new roots
only
root
grows
potential
transpiration
rate
= P = 2 bars
D
stomata close
actual transpiration rate
= P = 0 bars
-ve
Leaf water potential, Ψ


C translocated out of leaves
grow pods and seeds if appropriate
C remaining
N supply/demand
C to nodules
fix N
store C
grow nodules
initiate nodules
return C to shoot
C to plant
leaf water potential
C to root
grow roots
store C
return C to shoot
C to shoot
grow vegetative organs
initiate pods & seeds
initiate branches
store C
(reduce photosynthesis)


NKS 5960
Observed
Simulated
Plant Height (cm)
Vegetative Stages
Reproductive Stages
2379 kg ha$^{-1}$
2620 kg ha$^{-1}$
Days After Emergence
0
10
20
30
40
50
60
70
0
2
4
6
8
10
12
14
0
2
4
6
8
10
150
200
250
300